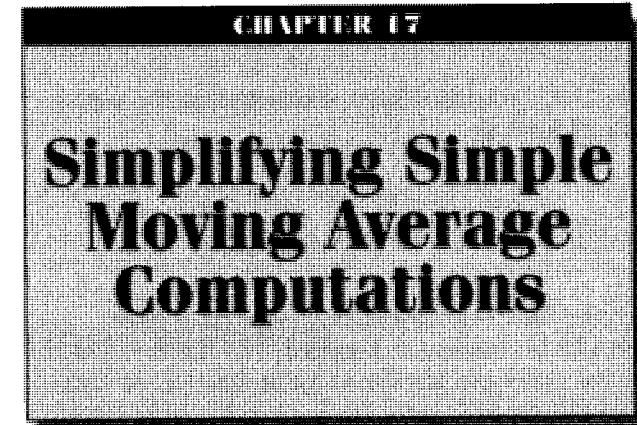


KEY POINTS TO REMEMBER

- Adding the difference between price and an exponential moving average to the price itself creates a leading indicator.
- The leading indicator always has noise gain.
- Smoothing the leading indicator with another exponential moving average can mitigate noise gain.
- Constants can be selected to provide a net lead for the indicator at low frequencies.
- The leading indicator has a lagging signal at price turning points.



"One topic has to be last," said Tom finally.

A simple moving average (SMA) of length N is computed by adding N values and dividing the sum by N . The process is repeated on a bar-by-bar basis. What could be easier? While conceptually easy, the coding for long moving averages can be tedious because there are so many terms. The tedium can be reduced by putting the summation in a loop. But looping is difficult to do in some applications, such as Excel. Another simplifying approach is to drop off the oldest value and add a new value to the moving average. But this requires computing the initial value of the long moving average at least once. I will show you two ways to compute the SMA with ease.

In Z transform notation, a unit delay is represented by Z^{-1} . The transfer response is the output of the filter divided by its input. Thus, the transfer response of an eight-bar SMA would be written as

$$H(z) = (1 + Z^{-1} + Z^{-2} + Z^{-3} + Z^{-4} + Z^{-5} + Z^{-6} + Z^{-7})/8 \quad (17.1)$$

This same expression, written in EasyLanguage where a delay of N bars is represented in square brackets as $[N]$, is shown in Equation 17.2.

$$\text{SMA} = (\text{Price} + \text{Price}[1] + \text{Price}[2] + \text{Price}[3] + \text{Price}[4] + \text{Price}[5] + \text{Price}[6] + \text{Price}[7])/8; \quad (17.2)$$

Equation 17.1 is a simple finite series that can be written most generally in fractional form as

$$H(z) = \frac{Y(z)}{X(z)} = \left(\frac{1 - z^{-N}}{1 - z^{-1}} \right) (N + 1) \quad (17.3)$$

where $Y(z)$ is the filter output and $X(z)$ is the filter input.

Equation 17.3 is identical to Equation 17.1 if $N = 7$, and is therefore an SMA. When we carry out the cross multiplication of Equation 17.3, we obtain

$$Y(z) = (X(z)(1 - z^{-N}) + Y(z)z^{-1})/(N + 1) \quad (17.4)$$

Equation 17.4 provides the means to program an arbitrarily long SMA using just a few terms. The EasyLanguage equivalent of Equation 17.4 is

$$\text{SMA} = (\text{Price} - \text{Price}[N] + \text{SMA}[1])/(N + 1); \quad (17.5)$$

Another SMA programming trick can be accomplished by recognizing that we don't have to do the filtering all at one time. Rather, we can cascade filters. That means we can filter the output of a previous filter that takes the output of a previous filter, and so on. Cascading filters are represented by multiplication in Z transforms. Therefore, the SMA transfer response of cascaded filters can be written as

$$H(z) = (1 + z^{-1})(1 + z^{-2})(1 + z^{-4}) \dots (1 + z^{-2^{K-1}})/2^K \quad (17.6)$$

For example, if $K = 3$, we would have an eight-bar SMA. As a test, we can expand Equation 17.6 to be

$$\begin{aligned} H(z) &= (1 + z^{-1})(1 + z^{-2})(1 + z^{-4})/8 \\ &= (1 + z^{-1} + z^{-2} + z^{-3})(1 + z^{-4})/8 \\ &= (1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6} + z^{-7})/8 \end{aligned} \quad (17.7)$$

Thus, Equation 17.7 shows that the cascaded filters expand to be identical with an SMA. In EasyLanguage, the cascaded filters would be written as

$$\begin{aligned} \text{Value1} &= \text{Price} + \text{Price}[1]; \\ \text{Value2} &= \text{Value1} + \text{Value1}[2]; \\ \text{Value3} &= \text{Value2} + \text{Value2}[4]; \\ \text{SMA} &= \text{Value3}/8; \end{aligned} \quad (17.8)$$

KEY POINTS TO REMEMBER

- An N -bar SMA can be written in an iterative form similar to an exponential moving average.
- An N -bar SMA can be written as K iterative two-element averages, where $N = 2^K$.